

PROJECT REPORT

1. Introduction

The Attendance Risk Predictor is a data-driven application designed to mitigate academic risk. While traditional systems simply record attendance, this project focuses on Predictive Analytics, using current data to forecast future outcomes. In an era where AI-driven academic intervention is becoming the standard, this tool serves as a foundational model for automated student support.

2. Problem Statement

The "75% Attendance Rule" creates significant stress for students due to:

- **Data Opacity:** Students lack a clear "path to recovery" once they fall below the threshold.
- **Decision Fatigue:** Constant manual calculations interfere with academic focus.
- **Late Detection:** Identifying an attendance crisis only at the end of a semester makes it mathematically impossible to fix.

3. Objectives

- To implement a predictive algorithm that identifies student "Risk Zones."
- To bridge the gap between Static Reporting and Dynamic Forecasting.
- To create a scalable logic structure that can eventually be trained on larger datasets (ML Integration).

4. Why This Project Matters

This project matters because it moves the student from a passive role to a Data Scientist role within their own education. It demonstrates the use of "Computational Thinking", breaking down a complex university regulation into a series of logical, repeatable steps.

5. Scope of the Project

- **Standardized Modelling:** Built for a 60-class semester (expandable).
- **Predictive Logic:** Not just calculating the past, but simulating the future.
- **Behavioural Advice:** Generating "Actionable Intelligence" for the user.

6. Methodology

The project utilizes a **Supervised Logic Approach**:

1. **Feature Input:** Total Classes, Classes Attended, and Classes Held.
2. **Target Variable:** The "45-Class Safety Mark" (75%).

3. **Simulation:** The program runs a "Best Case Scenario" simulation (assuming 100% future attendance) to check if the target is reachable.
4. **Classification:** Outputting the user into one of three classes: **Safe, At-Risk, or Critical.**

7. Implementation

The code is written in Python, utilizing its strength in data handling and clear syntax. It uses conditional structures to simulate a decision tree, which is the precursor to AI models like Random Forest.

8. AI/ML Relevance

This project is the first stage of an AI Pipeline:

- **Data Collection:** This script collects the necessary features for a dataset.
- **Feature Engineering:** Calculating "Max Possible Attendance" is a key feature that an ML model would use.
- **The ML Evolution:** A true ML version of this would take this data and add features like distance from campus and mid-term score to predict risk using a Logistic Regression or SVM (Support Vector Machine) model.

9. Challenges Faced

- **The "Impossible" Threshold:** Writing logic that accurately determines the point of no return required rigorous testing of edge cases.
- **User Interface:** Simplifying a mathematical forecast into easy-to-read "Advice" for the student.

10. Results

The model effectively categorizes students based on their mathematical probability of success. It provides an immediate "Reality Check," which is the primary goal of any predictive analytic tool.

11. Limitations

- **Deterministic:** It follows fixed rules and cannot yet "learn" from student behaviour patterns.
- **Data Isolation:** It does not account for external factors like health or extracurriculars that an AI would consider.

12. Future Enhancements

- **Regression Analysis:** Use ML to predict the *exact* final percentage a student will likely hit based on their current "rate of attendance."
- **Neural Networks:** Implementing a deep learning model to analyze attendance across 4 years of data to find long-term patterns.

- **Automated Data Scraper:** Pulling data directly from university portals using BeautifulSoup or Selenium.

13. What I Learned

I learned that the heart of AI/ML is not just complex math, but Solid Logic. Before a machine can learn, a human must define the rules of the problem. This project taught me how to translate a real-world problem into a dataset and a functioning algorithm.

14. Conclusion

The Attendance Risk Predictor is a successful prototype of a student-centric AI tool. It proves that with simple data and clear logic, we can significantly reduce academic risk and provide students with the clarity they need to succeed in a digital-first educational system.